Example for tautology

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Table 12.11

Since the last column of $p \vee \neg p$ contains only T, $p \vee \neg p$ is a tautology.

Example for contradiction

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Table 12.12

Since the last column contains only F, $p \wedge \neg p$ is a contradiction.

Note

All the entries in the last column of Table 12.10 are F and hence $(p \bar{\vee} q) \wedge (p \bar{\vee} \neg q)$ is a contradiction.

Example for contingency

p	q	$p \leftrightarrow q$	$\neg q$	$p \rightarrow \neg q$	$\neg(p \rightarrow \neg q)$	$(p \leftrightarrow q) \wedge \neg(p \rightarrow \neg q)$
T	T	T	F	F	T	T
T	F	F	T	T	F	F
F	T	F	F	T	F	F
F	F	T	T	T	F	F

Table 12.13

In the above truth table, the entries in the last column are a combination of T and F. The given statement is neither a tautology nor a contradiction. It is a contingency.

12.3.4 Duality

Definition 12.19

The **dual** of a statement formula is obtained by replacing $\vee$ by $\wedge$, $\wedge$ by $\vee$, T by F F by T. A dual is obtained by replacing $\mathbb{T}$ **(tautology)** by $\mathbb{F}$ **(contradiction)**, and, $\mathbb{F}$ by $\mathbb{T}$.

Remarks

(1) The symbol $\neg$ is not changed while finding the dual.

(2) Dual of a dual is the statement itself.

(3) The special statements $\mathbb{T}$ (tautology) and $\mathbb{F}$ (contradiction) are duals of each other.

(4) T is changed to F and vice-versa.

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